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Combating Counterfeit Products: Anti-counterfeiting Technology and Law Enforcement

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Abstract

Counterfeits cause tremendous damages to brand companies by eroding their market shares and reducing customers' buying motives. In response to such damages, many companies choose to develop and deploy anti-counterfeiting technologies, thus preventing counterfeiters from imitating their products. The companies can also rely on law enforcement exerted by local regulatory authorities to outlaw counterfeits. In this article, we study how brand companies may leverage the two approaches effectively. Specifically, we consider an authentic company that sells its products to a market that is administered by its regulatory authority. A counterfeiter exists in the market and may imitate the product offered by the authentic company. To combat counterfeiting, the authentic company may invest in anti-counterfeiting technology with uncertain outcomes, aiming to help customers effectively distinguish between genuine and fake products. We show that an increase in the authentic company's anti-counterfeiting capability does not necessarily yield a higher profit. We also find that, in equilibrium, the anti-counterfeiting effort exerted by the authentic company does not necessarily decrease with the penalty levied by the authority, even though they serve the same purpose. Our analysis further highlights the importance of coupling law enforcement type with anti-counterfeiting capability in the cooperation with the regularity authority. Specifically, when having a high (low, respectively) anti-counterfeiting capability, the authentic company should promote a cooperation that focuses on preventive (corrective, respectively) law enforcement.

Keywords

Deceptive Counterfeits, Anti-counterfeiting Technology, Law Enforcement, Game Theory

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1 Introduction

Counterfeit products have posed a severe threat to the global economy. According to the Organization for Economic Cooperation and Development (OECD), transactions of counterfeit and pirated goods accounted for up to 2.5% of global trade in 2019 (OECD, 2021). The U.S. Customs and Border Protection (CBP) reported that the monetary value of confiscated fakes have risen to $3.3 billion.[1] Alongside the increasing volume of counterfeit products, the counterfeiting business is becoming more diversified. Counterfeiters are no longer limited to producing clothing or footwear of well-known brands in labor-intensive industries. They have expanded their scopes into more sophisticated products, including electronics products (such as mobile phones) or healthcare products (such as drugs and vaccines). What makes the situation even worse is that counterfeiters are establishing new sales channels, including fake franchise stores and online shops on e-marketplaces in emerging economies (Asian Age, 2016). Counterfeiting activities cause substantial losses on brand companies. The presence of counterfeits erodes market share of genuine products. Moreover, counterfeits are typically of inferior quality

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compared to genuine ones. For consumers, the risk of getting counterfeits and thus suffering inferior qualities makes them reluctant to purchase. Decreased market shares and diluted purchase motives may in turn discourage brand companies from improving product quality and expanding product offerings, ultimately harming consumer welfare.

In practice, two approaches have been widely employed to prevent counterfeiting activities, legal actions and anti-counterfeiting technologies. Regulatory authorities in most countries have established Intellectual Property Rights (IPR) enforcement teams to identify and penalize IPR violations. Parallel to other practices of law enforcement, IPR enforcement teams can undertake either preventive actions or corrective actions. For the former, the IPR teams may actively inspect factory floors of local business owners. Whenever any counterfeiting activity is detected, the team suspends business licenses of related parties and confiscates equipment used for production. Via such preventive actions, the IPR enforcement teams can prevent counterfeit products from being produced and sold to consumers. On the other hand, if those products have reached the hands of consumers, the teams can turn to corrective actions. Specifically, in response to infringement reports about the selling of counterfeit products, the teams will collect evidence of the selling, freeze generated financial flows, engage in legal procedures of copyright or trademark violations, and compensate brand companies for their losses. That said, it is worth noting that IPR enforcement in emerging economies is indeed weak and limited due to large market sizes, inadequate budgets, and the focus placed on economic development. In fact, as reported by the media, 50% of the goods listed on the largest online and mobile marketplace in China are either fakes or products that infringe the IPR of others (Dasgupta, 2015).

Realizing the inefficiencies of IPR enforcement in emerging economies, many brand companies have turned to advanced anti-counterfeiting technologies to protect their products from being imitated. These brands develop and deploy anti-counterfeiting features (including but not limited to authenticity labels, certificates, and packages), which presumably may help their customers distinguish genuine products from counterfeit ones. According to a market report, the global anti-counterfeiting packaging market is projected to reach USD 326.3 billion by 2029.[2] Among these solutions, holographic anti-counterfeiting labels are the most well-known and widely used, particularly in the food and pharmaceutical industries (Haleem et al., 2022; Yang et al., 2024). Recently, leading pharmaceutical companies such as Pfizer took further steps to strengthen their efforts against counterfeit drugs by investing in more advanced technologies—including radio-frequency identification (RFID) and blockchain—to verify product authenticity in markets prone to counterfeiting (Pfizer, 2024).

Ironically, advanced technological developments such as 3D printing, digital scanning, and AI-driven design tools not only benefit brand companies but also enable counterfeiters to produce highly convincing fake products capable of deceiving customers (Meraviglia, 2018; Origify, 2024). This trend is further fueled by the presence of black markets in emerging economies, where illicit manufacturers are enabled to procure replicas of anti-counterfeiting features (Chen, 2023; Reuters, 2024a; Xinhua, 2016). Entities in black markets may reverse-engineer genuine products to replicate anti-counterfeiting features through various technological means. They may also bribe unethical employees of authentic companies to collect technical specifications or designs of proprietary anti-counterfeiting features, and then imitate such features with their own equipment. As reported by Wired (2007), while holograms were once considered a highly reliable method for authentication and were widely adopted across industries, counterfeiters can now easily order duplicates of holographic labels from suppliers via illicit channels.

As commented by GSK (2024), "[t]echnological advances have meant it is easier for criminals to accurately reproduce packaging[,] making it increasingly difficult for consumers to distinguish between authentic and counterfeit products." As a matter of fact, it has been well recognized by industrial practitioners that anti-counterfeiting technologies employed by pharmaceutical companies are likely to become accessible to counterfeiters within 12 to 18 months and need to be improved and updated regularly (Bansal et al., 2013). In short, while anti-counterfeiting technologies are widely adopted, it is not rare that such efforts could become unreliable. As a result, they may fail to boost either consumers' buying motives or brand companies' profits. Given the substantial costs associated with anti-counterfeiting measures and the potential unreliability of these technologies in combating counterfeiting activities, brand companies must carefully evaluate whether to invest in developing anti-counterfeiting technology and, if so, determine the appropriate level of investment (Reuters, 2024b).

Building on the above industry dynamics, this article seeks to answer the following research questions: Whether and when should an authentic company deploy an anti-counterfeiting technology, given the possibility that the technology developed is unreliable? How does the regulatory authority of the market affect the authentic company's decision on its anti-counterfeiting effort and its profit? What are the implications for the company if the regulatory authority places its focus on preventive enforcement, as opposed to corrective enforcement?

To address these research questions, we consider an authentic company that offers its products to a consumer market, where a counterfeiter may imitate the genuine products and sell them to consumers. To combat counterfeiting, together with its product offerings, the authentic company can exert an anti-counterfeiting effort to develop an anti-counterfeiting technology, preventing its products from being imitated. Meanwhile, the counterfeiter may acquire replicas of imitated anti-counterfeiting features from external sources and attach them to their products to imitate genuine products.

The cost of obtaining a replica depends on the reliability of the anti-counterfeiting technology developed by the authentic company. If the anti-counterfeiting technology turns out to be reliable (unreliable, respectively), the cost of a replica for the counterfeiter is high (low, respectively). The authentic company's anti-counterfeiting effort determines the probability of the anti-counterfeiting technology being reliable (the larger the investment, the more likely the technology turns out to be reliable).

Conditional on the realized imitation cost, the counterfeiter decides on whether or not to make deceptive counterfeits. If the counterfeiter decides to do so, it faces a penalty levied by the regulatory authority of the market once the counterfeiting activity is detected either before or after the sales. For customers in the market, they cannot differentiate between genuine products offered by the authentic company and deceptive counterfeits made by the counterfeiter (for customers, they cannot distinguish two products based on their appearance including some anti-counterfeiting features). Furthermore, they have no information regarding the decision made by the authentic company on exerting the anti-counterfeiting effort and that by the counterfeiter on producing the fake. As a result, she has to form a belief when making the purchase decision.

We find that the authentic company's equilibrium profit is not monotonically increasing in its anti-counterfeiting capability. In more detail, when the company's anti-counterfeiting capability is relatively low, a marginal increase in its capability may backfire by reducing its equilibrium profit. This occurs because the increase in the anti-counterfeiting capability makes the counterfeiter more likely to produce and sell a fake, thereby completely negating the benefit gained from the increased capability. This result suggests that for a brand company targeting high profitability, a significant improvement in its anti-counterfeiting capability can be necessary. We also find that the anti-counterfeiting effort exerted by the authentic company does not necessarily decrease in the penalty levied by the regulatory authority, even though they serve the same purpose of countering the counterfeiting activity. In fact, an increase in the penalty has two opposite effects on the company's profit. On the one hand, it can discourage the counterfeiter from making and selling its fake, thus helping save the anti-counterfeiting cost; on the other hand, it can attract more customers to make a purchase, which drives the authentic company to exert more effort. Our analysis shows that the latter effect dominates when the penalty is relatively light and the counterfeiter does not enter the market with certainty (i.e., the counterfeiter plays mixed strategy). This result can yield an important and non-intuitive message to brand companies that engage in a business relationship with emerging economies. Specifically, while some governments of those economies have started to strengthen their IPR enforcement (e.g., by charging higher penalties on counterfeiting activities), it probably cannot help decrease brand companies' spendings on anti-counterfeiting technologies because the companies may hardly enjoy a free ride by relying purely on the improved enforcement. Instead, they need to increase their investments in anti-counterfeiting technologies as well. In addition, we show that when the authentic company's anti-counterfeiting capability is high, increasing corrective reinforcement intensity could disincentivize the company from exerting its effort to protect its products from being imitated, which could inadvertently lead to an increased probability of counterfeit occurrences.

We further analyze the preventive and corrective law enforcements and compare their impacts. Our analysis reveals that compared with corrective law enforcement (which penalizes the counterfeiter after the selling of its products), when the anti-counterfeiting capability is low, preventive law enforcement (which outlaws the counterfeiter beforehand) may reduce the counterfeit occurrence probability in equilibrium more effectively, at the cost of bringing a lower equilibrium profit to the company. The reason comes from an over-investment problem. In fact, preventive law enforcement raises customers' expectations for the product quality and thus motivates the counterfeiter to enter the market. This forces the authentic company to increase its anti-counterfeiting effort as well, thereby raising the company's cost burden. This finding highlights the need for brand companies, including those with strong anti-counterfeiting capabilities, to maintain close collaboration with regulatory authorities. It also underscores the importance of the coupling between the company's anti-counterfeiting capability and the regulatory authority' law enforcement type: a company with a high (or low, respectively) anti-counterfeiting capability should advocate for a cooperation that emphasizes preventive (or corrective, respectively) law enforcement. We also study the consumer welfare implications of these two law enforcement types and show that with a high anti-counterfeiting capability being in place, preventative law enforcement benefits both the authentic company and customers.

The remainder of the article is organized as follows: in Section 2, we review related literature. We then present the model and assumptions in Section 3 and derive the equilibrium solution in Section 4. In Section 5, we examine the impact of the regulatory authority's law enforcement on the authentic company's equilibrium strategy, followed by some extensions and discussions in Section 6. Finally, we conclude our article in Section 7. All the proofs and extended discussions can be found in the E-Companions.

2 Literature Review

While Leibenstein (1950) studied the impact of counterfeit products on consumer behavior, early works on counterfeit products either focus on photocopying problems (Besen and Kirby, 1989; Johnson, 1985; Liebowitz, 1985) or are qualitative in nature (Harvey, 1987; Harvey and Ronkainen, 1985).

Grossman and Shapiro (1988a, 1988b) initiated a stream of literature that models and analyzes the trade problem of counterfeit products. They classified counterfeit products into

two types, non-deceptive counterfeit products and deceptive ones. For non-deceptive counterfeit products, consumers know that they are purchasing counterfeit products (as opposed to genuine products) after a close inspection or by inferring information from product distribution channels. Thus, in this scenario, consumers choose to buy counterfeit products mainly for the sake of low price. Economic impacts of non-deceptive counterfeit products have been extensively studied, with representative references including Grossman and Shapiro (1988a), Conner and Rumelt (1991), Banerjee (2003), Yao (2005), Tsai and Chiou (2012), Gao et al. (2017), Pun and DeYong (2017), Yi et al. (2022), Chen et al. (2022), and Lu et al. (2023). For deceptive counterfeit products, consumers are incapable of distinguishing them from genuine goods. Hence, the consumers can at best form a belief about the genuineness of products available in the market and then make their purchase decisions based on that belief.

In research on deceptive counterfeit products, the majority of studies focus on how genuine companies deter counterfeiters from entering the market. Grossman and Shapiro (1988b) studied an international trade problem where foreign suppliers produce both legitimate low-quality products and counterfeit products that imitate genuine products offered to the domestic market. Zhang and Zhang (2015) developed an optimal supply chain structure in a market with deceptive counterfeit products. Cho et al. (2015) demonstrated that the effectiveness of anti-counterfeiting strategies depends largely on whether an authentic company faces a non-deceptive counterfeiter or a deceptive one. Ding (2014) found that the presence of counterfeit products helps a monopolist alleviate the time-inconsistency problem. Based on a vertical product differentiation model, Qian (2014) took into account two layers of asymmetric information and shows that enforcement and fixed-cost signals can relax conditions for separating equilibrium. Qian et al. (2015) showed that an authentic company tends to improve its searchable quality more than its experiential quality. Gao (2018) modeled overt technologies as a fixed barrier cost for counterfeiters and examine how this barrier cost affects counterfeiter entry into a dubious drug market. Pun et al. (2021) explored the value of adopting the blockchain technology and how the adoption may be influenced by government policies, including subsidies and corrective enforcement. Shen et al. (2022) illustrated that the usage of a permissioned blockchain technology benefits a brand company (i.e., the incumbent) via discouraging the entry of a copycat (i.e., the counterfeiter) and thus affects the supply chain performance of the company. Gao and Wu (2023) examined a retailer's decision on whether to sell authentic products or counterfeit goods, and how regulation influences this choice. Lu et al. (2023) studied how converting counterfeiters can benefit brand companies and influence the social welfare. Li et al. (2023) addressed a product authenticity problem in which a platform offers inspection services to buyers.

As we consider a problem in which customers cannot differentiate a genuine product made by an authentic company from a fake by a counterfeiter, this article is in line with the stream of literature that studies deceptive counterfeits. Our game-theoretical model differs from the existing literature (e.g., Pun et al. 2021) via the following three salient features: First, while the existing literature assumes that the decision on adopting the anti-counterfeiting technology is binary and once adopted, it yields a certain outcome, our paper treats the investment in the anti-counterfeiting technology as a continuous decision variable associated with uncertain outcomes; Second, in our model, the authentic company cannot commit its anti-counterfeiting effort to customers, which reflects the reality that consumers often lack the domain knowledge to assess the effort and have to form a prior belief without relying on it. Third, to the best of our knowledge, this work is the first study to consider both preventive and corrective law enforcements in the same modeling framework and compare their impacts on the optimal anti-counterfeiting strategy. These modeling features set a distinction between our paper and those in the literature and render rather unique results and insights.

3 Model and Assumptions

Consider an authentic company that sells its products. The unit production cost is k and the unit price is $p > k$.[3] A counterfeiter exists in the market and may produce fake products that imitate the genuine ones produced by the authentic company but is of an inferior quality.

The mass of customers in the market is normalized to one. We assume that customers are heterogeneous regarding their willingness to pay. We use a parameter θ to capture this heterogeneity and assume that θ is uniformly distributed between 0 and 1. If a customer with $\theta \in (0, 1)$ purchases a genuine product at price p, she gains a net utility of $\left(\frac{q}{\theta}\right)^{\frac{1}{\gamma}} - p$, with $q > 0$ and $\gamma > 1$. If a customer with $\theta \in (0, 1)$ purchases a fake product at price p, she gains a net utility of $-p$. It can be shown that the induced demand for the genuine products takes the form of q/p^{γ}, which is a demand function with constant elasticity γ often studied in the economics and operations management literature.[4] In EC.2, we will demonstrate the robustness of our results under a different consumer utility function.

For a customer, she is not able to tell the genuineness of a product at the time when she decides on whether to purchase a unit.[5] As a result, the type-θ customer can at most form a belief $b(\theta)$ about the genuineness of the product she encounters on the market, and then makes the purchase decision based on her belief. Given the belief, the customer will buy a product if and only if he gains a non-negative utility, that is, $b(\theta)\left(\frac{q}{\theta}\right)^{\frac{1}{\gamma}} - p \geq 0$. We use the indicator function $\delta(\theta)$ to represent the purchase decision made by the type-θ customer. Specifically, $\delta(\theta) = 1$ if she buys and $\delta(\theta) = 0$ if she does not buy.

To combat counterfeiting, the authentic company can invest in and develop an anti-counterfeiting technology. However, as highlighted in the introduction, the technology may not

be perfect. After the company launches its genuine products, the counterfeiter may obtain imitated anti-counterfeiting features (e.g., labels, tags, certificates, or packages) through external channels such as black markets.[6] In what follows, we will refer to these anti-counterfeiting features collectively as "labels," though the scope extends beyond its conventional meaning. If the authentic company invests more in its anti-counterfeiting technology, the resulting features become technologically more difficult for the black markets to imitate. Thus, it is more likely that the anti-counterfeiting technology will turn out to be reliable, making it extremely costly for the counterfeiter to obtain the imitation labels.

Formally, we let $e \in [0, 1]$ be the probability that the anti-counterfeiting technology is reliable. A reliable (unreliable, respectively) anti-counterfeiting technology yields a high (low, respectively) product cost for the counterfeiter (which comprises the counterfeiter's manufacturing expenses[7] and the cost of obtaining a label replica), that is, $c = c_h$ ($c = c_l$, respectively). To sharpen our insight, we assume $c_h = +\infty$, meaning that the higher unit cost of imitation label makes it unprofitable for the counterfeiter to produce and sell its product (alternatively, we can interpret this case as imitation labels being unavailable for purchase through external channels, making it impossible for the counterfeiter to engage in any deceptive counterfeiting business).[8] Since the parameter e affects the reliability of the anti-counterfeiting technology, we refer to the probability e as the anti-counterfeiting effort. Note that the anti-counterfeiting effort modeled here can also be extended to include the effort put into safeguarding the technology developed (from any illegal sales to black markets).

We denote by $\frac{1}{t}\psi(e)$ the disutility associated with the anti-counterfeiting effort, where $t > 0$ is the cost efficiency of developing an anti-counterfeiting technology.[9] A larger t represents a lower cost for exerting the same anti-counterfeiting effort level e, reflecting the authentic company's anti-counterfeiting capability. We assume that $\psi(e)$ takes the form of $e^n/(1-e)$ with $n \geq 2 + \gamma$, which is a convex function[10] in e and goes to infinity as e approaches to 1. Such modeling assumption enables us to avoid boundary solutions and captures the fact that brand companies need to increase their technological investments in order to reduce the occurrence of counterfeiting activities. Note that similar assumptions on disutility functions are rather common in the literature; see, for instance, Baiman et al. (2000).

If the anti-counterfeiting technology turns out to be unreliable (i.e., $c = c_l$), the counterfeiter may choose to produce and sell its products or not (i.e., entry decision). We denote by $s \in [0, 1]$ the probability that the counterfeiter with $c = c_l$ makes and sells its products. The inclusion of a mixed strategy for the counterfeiter in our game analysis serves two purposes: first, it ensures the existence of an equilibrium (see the detailed discussion in Section 4); second, it introduces unpredictability in counterfeiting activities, making it more challenging for the authentic company to determine its optimal anti-counterfeiting effort. Later, we will demonstrate that a mixed strategy equilibrium emerges within a specific parameters region (also see Babich and Tang (2012) for an analysis of mixed strategy equilibrium).

The availability of counterfeit products could affect the sale of the genuine products to different degrees. For example, the adverse impact could be low (high, respectively) if the authentic company (the counterfeiter, respectively) can effectively guide customers to visit their stores. In this stylized model, to emphasize the effect of law enforcements on combating counterfeit products, we consider the worst case. That is to say, the sales of the genuine products are zero if the counterfeiter is able to sell goods on the market (we will relax this assumption in Section 6.2).

Selling counterfeit products is illegal in this economy. To hinder such illegal activities and protect the IPR, the regulatory authority of the market forms an enforcement team to administer the market. The team can put factory floors into close scrutiny. As a result, the counterfeiter may be detected, followed immediately by a *preventive* legal action, such as a revocation of its business license and a confiscation of its production equipment prior to the occurrence of any sale. We assume that the enforcement of the preventive legal action takes place with a probability of $\alpha_1 \in [0, 1]$. Then, given that the counterfeiter is not caught beforehand, it may be discovered later by the authentic company after the selling of the fake products. Specifically, upon receiving a violation report from the authentic company, the enforcement team follows up by collecting evidence and engaging in a prosecution against the counterfeiter. We assume that the enforcement of such *corrective* legal action occurs with a probability of $\alpha_2 \in [0, 1]$. Regardless of how the counterfeiter is identified, via the preventive legal action or the corrective one, the financial penalty imposed on the counterfeiter is L, as a fine for conducting the illegal business. Besides, with the corrective legal action being in place, the counterfeiter's illegal gain will be seized, part of which will be used to compensate the authentic company's profit loss via an IPR infringement lawsuit (see Chow (2020), for reference).

The sequence of events is as follows:

(1) The authentic company exerts its anti-counterfeiting effort level e. Then, whether the anti-counterfeiting technology is reliable or not is realized.
(2) If the anti-counterfeiting technology turns out to be unreliable (that is, $c = c_l$), the counterfeiter makes the entry decision s.
(3) Whether the preventive legal action is enforced is realized.
(4) If the preventive legal action is in place, the counterfeiter incurs the penalty cost L.
(5) The customers decide on whether to buy a product from the market.
(6) Whether the corrective legal action is enforced is realized.

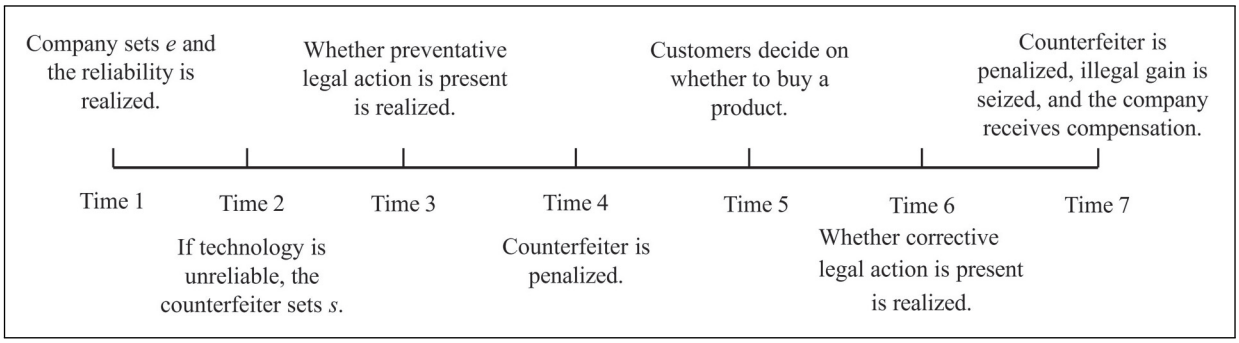


Figure 1. Sequence of events.

(7) If the corrective legal action is in place, the counterfeiter incurs the penalty cost L, its illegal gain is seized, and the authentic company receives a compensation for its profit loss. See Figure 1 for a graphical illustration of the sequence of events.

In the base model setting, the customers, when deciding whether to buy a product on the market, cannot observe the authentic company's anti-counterfeiting effort e and its realization on the counterfeiter's product cost, and the counterfeiter's entry decision s due to lack of domain knowledge.[11] Due to a short presence on the market, the customers cannot observe the realizations of legal actions. In addition, to focus on the effects of the decisions made by the authentic company and the counterfeiter, and to maintain tractability, we assume that all model parameters and cost functions are common knowledge.

To recap, our model has three unique features that set it apart from the existing literature. First, it treats the investment in the anti-counterfeiting technology as a continuous decision variable associated with uncertain outcomes. Second, it assumes that the authentic company's anti-counterfeiting effort is non-committal to customers due to their lack of domain knowledge. Third, it incorporates and compares both preventative and corrective law enforcement measures in the same modeling framework. As will be shown later, those features contribute to generating rather unique results and insights.

4 Equilibrium Formulation and Analysis

In this section, we first establish the authentic company's and counterfeiter's objective functions. Second, we define the equilibrium for this game. Third, we derive the equilibrium solution for this game.

First, the authentic company's profit function can be formulated as follows:

$$\begin{aligned}&\Pi_a\left(e;s,\{\delta(\theta)\}_{\theta\in(0,1)}\right)\\&=\left[e+(1-e)\left(1-s+s\alpha_1\right)\right](p-k)\int_0^1\delta(\theta)\,d\theta\\&\quad+(1-e)s\left(1-\alpha_1\right)\alpha_2(p-k)\int_0^1\delta(\theta)\,d\theta-\frac{1}{t}\psi(e),\end{aligned}\tag{1}$$

where the first term $[e+(1-e)(1-s+s\alpha_1)](p-k)\int_0^1\delta(\theta)d\theta$ represents the authentic company's expected profit from the market, the second term $(1-e)s(1-\alpha_1)\alpha_2(p-k)\int_0^1\delta(\theta)d\theta$ stands for the expected compensation via the corrective law enforcement, and the last term $\frac{1}{t}\psi(e)$ denotes the disutility of exerting the anti-counterfeiting effort.

Next, we can formulate the profit function of the counterfeiter with $c=c_l$ as follows (note that when $c=c_h$, the counterfeiter never enters the market, so its profit always equals zero):

$$\begin{aligned}&\Pi_s\left(s;\{\delta(\theta)\}_{\theta\in(0,1)}\right)\\&=\left(1-\alpha_1\right)sp\int_0^1\delta(\theta)\,d\theta-\left(1-\alpha_1\right)\alpha_2 sp\int_0^1\delta(\theta)\,d\theta\\&\quad-(1-\alpha_1)c_l s\int_0^1\delta(\theta)\,d\theta-\alpha_1 Ls-\left(1-\alpha_1\right)\alpha_2 Ls,\end{aligned}\tag{2}$$

where the first term $(1-\alpha_1)sp\int_0^1\delta(\theta)d\theta$ is the expected revenue gained from producing and selling counterfeits, the second term $(1-\alpha_1)\alpha_2 sp\int_0^1\delta(\theta)d\theta$ is the expected illegal gain seized by the corrective law enforcement, the third term $(1-\alpha_1)c_l s\int_0^1\delta(\theta)d\theta$ is the expected product cost, and the fourth term $\alpha_1 Ls$ and the fifth term $(1-\alpha_1)\alpha_2 Ls$ represent the expected penalties levied by the authority via the preventive and corrective law enforcement, respectively.

With the above preparation, we are in position to define the equilibrium solution in this game. We adopt the concept of perfect Bayesian equilibrium throughout the article.

DEFINITION 1. *A profile of strategies* $(e^*, s^*, \{\delta^*(\theta)\}_{\theta\in(0,1)})$ *and belief* $\{b^*(\theta)\}_{\theta\in(0,1)}$ *is an equilibrium if the following conditions hold:*

(i) $e^* \in \arg\max_e \Pi_a\left(e; s^*; \{\delta^*(\theta)\}_{\theta\in(0,1)}\right)$,
(ii) $s^* \in \arg\max_s \Pi_s\left(s; \{\delta^*(\theta)\}_{\theta\in(0,1)}\right)$,
(iii) $\delta^*(\theta) = \begin{cases} 1 & \text{if } b^*(\theta)\left(\frac{q}{\theta}\right)^{\frac{1}{\gamma}} \geq p \\ 0 & \text{if } b^*(\theta)\left(\frac{q}{\theta}\right)^{\frac{1}{\gamma}} < p \end{cases}$ *for all* $\theta \in (0,1)$,
(iv) $b^*(\theta) = 1 - (1-e^*)(1-\alpha_1)s^*$ *for all* $\theta \in (0,1)$.

Conditions (i) and (ii) require that both the authentic company and the counterfeiter maximize their respective profits in equilibrium. Condition (iii) means that the customers maximize their utility based on their belief. Finally, condition (iv) mandates that the customers' equilibrium belief be consistent with equilibrium play. We note from condition (iv) that $b^*(\theta)$ is constant with respect to θ. Hence, we now use b^* to represent $b^*(\theta)$. With the constant belief function, the customers' purchase decision simply follows a threshold policy at equilibrium: $\delta^*(\theta) = 1$ if $\theta \leq q\left(\frac{b^*}{p}\right)^{\gamma}$ and $\delta^*(\theta) = 0$ if $\theta > q\left(\frac{b^*}{p}\right)^{\gamma}$.

Applying the first-order condition to the authentic company's problem yields the best response associated with the anti-counterfeiting effort as follows:

$$\frac{1}{t}\psi'(e^*) = s\left(1-\alpha_1\right)\left(1-\alpha_2\right)(p-k)\int_0^1 \delta^*(\theta)\,d\theta. \quad (3)$$

As demonstrated in (3), the authentic company's best response balances the marginal investment in anti-counterfeiting, $\frac{1}{t}\psi'(e)$, and the total marginal benefit gained, $s(1-\alpha_1)(1-\alpha_2)(p-k)\int_0^1 \delta^*(\theta)d\theta$. It confirms the intuition that a low probability of the counterfeiter entering, a high probability of the counterfeiter being identified, and a low customer demand all dilute the authentic company's incentive to exert its anti-counterfeiting effort.

Similarly, by solving the counterfeiter's optimization problem, we obtain that the counterfeiter's best response strategy satisfies

$$s^* \in \begin{cases} \{1\} & \text{if } L < \frac{(1-\alpha_1)(1-\alpha_2)p-(1-\alpha_1)c_l}{\alpha_1+(1-\alpha_1)\alpha_2}\int_0^1 \delta^*(\theta)\,d\theta, \\ [0,1] & \text{if } L = \frac{(1-\alpha_1)(1-\alpha_2)p-(1-\alpha_1)c_l}{\alpha_1+(1-\alpha_1)\alpha_2}\int_0^1 \delta^*(\theta)\,d\theta, \\ \{0\} & \text{if } L > \frac{(1-\alpha_1)(1-\alpha_2)p-(1-\alpha_1)c_l}{\alpha_1+(1-\alpha_1)\alpha_2}\int_0^1 \delta^*(\theta)\,d\theta. \end{cases} \quad (4)$$

As is clear from the formulation, there exist four drivers that motivate the counterfeiter to make the counterfeits, including a high customer demand, a small probability of being caught, a low penalty once caught, and a low counterfeit product cost. To focus on the impact of law enforcements and customer demand on the counterfeiter's decision and to make the expressions concise, we normalize c_l to zero, which can be reasonable for cases where the counterfeiter's labor and material costs are quite low (Chen, 2023; Xinhua, 2016) and is common in the literature (Cho et al., 2015; Pun et al., 2021; Qian, 2014). We relegate the discussion of a positive product cost to Section 6.1.

We let Π^* be the authentic company's equilibrium profit. We further denote by σ^* the equilibrium counterfeit occurrence probability, which can be calculated as follows:

$$\sigma^* = (1-e^*)(1-\alpha_1)s^*. \quad (5)$$

Note that $\sigma^* = 1 - b^*$. Hence, the equilibrium counterfeit occurrence probability reflects how attractive a product is on the consumer market.

For notational simplicity, we define $\alpha^\dagger = \frac{\alpha_1+(1-\alpha_1)\alpha_2}{(1-\alpha_1)(1-\alpha_2)}$. It is straightforward to see that $\alpha^\dagger$ is increasing in α_1 and α_2, thereby capturing the magnitude of law enforcement. With the above analysis and notation, we are ready to present the equilibrium outcome in the following proposition. The equilibrium customers' decisions are the same as the ones defined in Definition 1, which we omit for brevity.

PROPOSITION 1. *The equilibrium outcome can be characterized as follows:*

(a) *If* $L \geq \frac{p}{\alpha^\dagger}\min\left\{1, \frac{q}{p^\gamma}\right\}$, *the authentic company's anti-counterfeiting effort is* $e^* = 0$, *the counterfeiter's effort is* $s^* = 0$, *and the customers' belief is* $b^* = 1$; *moreover, the authentic company's equilibrium profit is* $\Pi^* = (p-k)\min\left\{1, \frac{q}{p^\gamma}\right\}$ *and the resulting counterfeit occurrence probability is* $\sigma^* = 0$.
(b) *If* $L < \frac{p}{\alpha^\dagger}\min\left\{1, \frac{q}{p^\gamma}\right\}$, *the counterfeiter's effort is* $s^* > 0$ *and there exists a threshold*

$$\bar{t} := \frac{\psi'\left(\left[\frac{p}{1-\alpha_1}\left(\frac{1}{p}\frac{\alpha^\dagger L}{q}\right)^{\frac{1}{\gamma}} - \frac{\alpha_1}{1-\alpha_1}\right]^+\right)}{(p-k)\frac{\alpha_1 L+(1-\alpha_1)\alpha_2 L}{p}},$$

such that:

(i) *when $t < \bar{t}$, e^* is the unique solution to*

$$\frac{(1-e)\psi'(e)}{t} = \left(1-\alpha_2\right)(p-k)\frac{\alpha^\dagger L}{p} \times \left(1-p\left(\frac{1}{q}\frac{\alpha^\dagger L}{p}\right)^{\frac{1}{\gamma}}\right),$$

$$s^* = \frac{1}{\left(1-\alpha_1\right)(1-e^*)}\left(1-p\left(\frac{1}{q}\frac{\alpha^\dagger L}{p}\right)^{\frac{1}{\gamma}}\right),$$

and $b^* = p\left(\frac{1}{q}\frac{\alpha^\dagger L}{p}\right)^{\frac{1}{\gamma}}$;

$$\Pi^* = (p-k)\frac{\alpha^\dagger L}{p}\left(\alpha_2 + (1-\alpha_2)p\left(\frac{1}{q}\frac{\alpha^\dagger L}{p}\right)^{\frac{1}{\gamma}}\right) - \frac{\psi(e^*)}{t}, \quad \textit{and} \quad \sigma^* = 1-p\left(\frac{1}{q}\frac{\alpha^\dagger L}{p}\right)^{\frac{1}{\gamma}}.$$

(ii) *when $t \geq \bar{t}$, e^* is the unique solution to*

$$\frac{1}{t}\psi'(e) = \left(1-\alpha_1\right)\left(1-\alpha_2\right)(p-k) \times \min\left\{q\left(\frac{e+(1-e)\alpha_1}{p}\right)^{\gamma}, 1\right\},$$

$s^* = 1$, *and* $b^* = e^* + (1-e^*)\alpha_1$;

$$\Pi^* = [e^* + (1-e^*)(\alpha_1 + (1-\alpha_1)\alpha_2)](p-k) \times \min\left\{q\left(\frac{e^*+(1-e^*)\alpha_1}{p}\right)^{\gamma}, 1\right\} - \frac{\psi(e^*)}{t}$$

and $\sigma^* = (1-e^*)(1-\alpha_1)$.

Proposition 1 confirms our intuition that whether the authentic company exerts its anti-counterfeiting effort depends on the penalty level L levied by the regulatory authority. When the penalty L is large, that is, $L \geq \frac{p}{\alpha^\dagger}\min\left\{1, \frac{q}{p^\gamma}\right\}$, the counterfeiter has no interest in imitating the product (i.e., $s^* = 0$), yielding the counterfeit-free case (i.e., $\sigma^* = 0$). In this case, there is no need for the authentic company to exert any anti-counterfeiting effort (i.e., $e^* = 0$) and the customers whose θ are less than $\min\left\{1, \frac{q}{p^\gamma}\right\}$ will purchase the product on the market.

When the penalty L is low, that is, $L < \frac{p}{\alpha^\dagger}\min\left\{1, \frac{q}{p^\gamma}\right\}$, the counterfeiter with low product cost (i.e., $c = c_l$) would venture into counterfeiting (i.e., $s^* > 0$). Depending on the anti-counterfeiting capability (i.e., the value of t), the counterfeiter may choose to enter the market with varying probabilities. When the anti-counterfeiting capability is relatively low, that is, $t < \bar{t}$, the counterfeiter plays a mixed strategy with a probability of $s^* \in (0,1)$ whenever imitating the genuine product is possible. That is, the counterfeiter may or may not enter into the market and gains an expected profit of zero in equilibrium. (Note that the other cases cannot constitute equilibrium. On the one hand, the counterfeiter cannot choose $s^* = 0$; otherwise, the market expectation is high enough to make it profitable for the counterfeiter to enter into the counterfeiting business. On the other hand, the counterfeiter cannot choose $s^* = 1$; otherwise, the market expectation is too low for the counterfeiter to gain a non-negative profit if entering.) Expecting that the counterfeiter is likely to make a fake, the authentic company chooses to develop and implement an anti-counterfeiting technology ($e^* > 0$). When the anti-counterfeiting capability is relatively high, that is $t > \bar{t}$, the authentic company has the high capability to prevent the counterfeiter from counterfeiting the products, yielding a high expectation on the product quality on the market. This makes the counterfeiter (if its realized product cost is low) to produce a fake with certainty ($s^* = 1$) to pursue high profitability. That way, it can earn a positive expected profit because the potential gain from the market outweighs the expected value of the penalty to be levied by the regulatory authority. Anticipating the counterfeiter's decision, the authentic company continues to exert the effort in this case.

With the characterization of the equilibrium solution, we are ready to study how the company's anti-counterfeiting capability affects the equilibrium strategy and the equilibrium profit of the authentic company. From Proposition 1, it is clear to see that when the penalty L is sufficiently large, that is, $L \geq \frac{p}{\alpha^\dagger}\min\left\{1, \frac{q}{p^\gamma}\right\}$, both e^* and σ^* are equal to zero, which is a trivial case. To make the discussion more interesting, we focus on the case of $L < \frac{p}{\alpha^\dagger}\min\left\{1, \frac{q}{p^\gamma}\right\}$.

PROPOSITION 2. *Suppose $L < \frac{p}{\alpha^\dagger}\min\left\{1, \frac{q}{p^\gamma}\right\}$. The following comparative statics results hold:*

(a) *Both the authentic company's equilibrium anti-counterfeiting effort level e^* is increasing in t; the counterfeiter's equilibrium effort level s^* is increasing in t for $t \in [\bar{t}, +\infty)$ and is constant for $t \in [\bar{t}, +\infty)$; the counterfeit occurrence probability σ^* is constant in t for $t \in (0, \bar{t})$ and decreasing in t for $t \in [\bar{t}, +\infty)$.*

(b) *The authentic company's equilibrium profit Π^* is quasi-convex in t. Specifically, Π^* is first decreasing in t for $t \in (0, \bar{t})$ and then increasing for $t \in [\bar{t}, +\infty)$.*

Proposition 2(a) shows that with increased anti-counterfeiting capability, the counterfeiter with $c = c_l$ becomes more motivated to produce and sell the counterfeit products. As a response, the authentic company has more incentive to exert anti-counterfeiting effort to protect its products from being imitated (see Figure 2(a) for an illustration).

More interestingly, Proposition 2(b) reveals the impact of the anti-counterfeiting capability on the equilibrium profit. As illustrated in Figure 3(a), the authentic company's equilibrium profit Π^* is non-monotone in t. When the capability is low, its

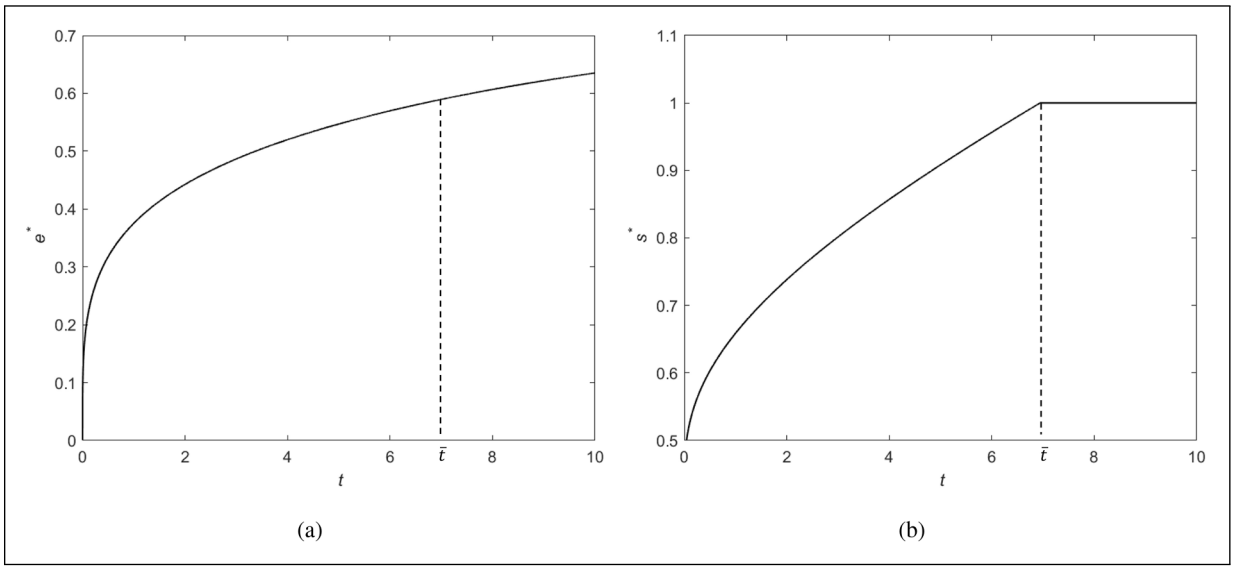


Figure 2. Equilibrium decisions in the base model; model parameters: $q = 10$, $p = 8$, $k = 2$, $L = 1$, $\alpha_1 = 0.2$, $\alpha_2 = 0.2$,$\gamma = 2$, and $n = 5$.

equilibrium profit Π^* is decreasing in t. In this case, improving the anti-counterfeiting capability makes the counterfeiter more likely to produce and sell the counterfeits (i.e., set a high value of s^*) when it is possible. Thus, from the customers' viewpoint, the benefit from the increased capability is completely wiped out by an increased possibility of getting a counterfeit product, yielding an invariance in customers' expectation of the genuineness of the product offered on the market (see the flat line segment in Figure 3(b)). This fact, together with the increased cost of exerting more anti-counterfeiting effort caused by the increased market share of the counterfeiter, leads to a lower equilibrium profit. When the anti-counterfeiting capability is high, that is, $t > \bar{t}$, the company's equilibrium profit Π^* is increasing in t. In this case, the counterfeiter enters the market with certainty (i.e., s^* reaches the maximum value). Therefore, an increase in the anti-counterfeiting capability attracts more demand, leading to a higher profit.

The structural property of Π^* illustrated above can yield an important and non-intuitive message to brand companies. Under the threat of counterfeiter entry, a lower anti-counterfeiting capability does not necessarily imply a lower profit for the authentic company. In fact, it could provide the company with a commitment device to mitigate fierce reactions from counterfeiters (see more discussions in Section 6.3). In short, with the potential entry of counterfeiters, if brand companies target increasing their profitability but own a relatively low anti-counterfeiting capability, they need to avoid attempting any marginal improvement on the anti-counterfeiting capability. Instead, they should seek opportunities to enhance the capability significantly. Alternatively, they can turn to strengthening their cooperation with regulatory authorities in outlawing counterfeits, as will be discussed next.

5 Implications of Law Enforcements

In this section, we study the impacts of law enforcement. We first investigate how the authentic company's equilibrium anti-counterfeiting effort is influenced by the penalty level L levied by the regulatory authority and two law enforcement intensities α_1 and α_2. Then, we compare the two law enforcements and discuss their implications.

5.1 Impact of Penalty and Enforcement Intensity

The following proposition presents the impact of penalty on the company's equilibrium effort and the resulting counterfeit occurrence probability.

PROPOSITION 3. *Suppose* $L < \frac{p}{\alpha^\dagger} \min\left\{1, \frac{q}{p^\gamma}\right\}$. *The comparative statics results for L are:*

(a) σ^* *is decreasing in L if* $t < \bar{t}$ *and constant in L otherwise;*

(b) e^* *is increasing in L when* $L < \left(\frac{\gamma}{(1+\gamma)p}\right)^\gamma \frac{qp}{\alpha^\dagger}$ *and* $t < \bar{t}$, *decreasing in L when* $L > \left(\frac{\gamma}{(1+\gamma)p}\right)^\gamma \frac{qp}{\alpha^\dagger}$ *and* $t < \bar{t}$, *and constant in L otherwise.*

The comparative statics for the equilibrium counterfeit occurrence probability is rather intuitive. As shown in Proposition 3, this probability is non-increasing in the penalty L, indicating that a stringent law enforcement in general can reduce the counterfeit occurrence probability. It is worth commenting that there exist two scenarios in which increasing the penalty would not help reduce the counterfeit occurrence probability. The first scenario is a trivial one and occurs when the counterfeiter does not participate in counterfeiting ($s^* = 0$) due to a

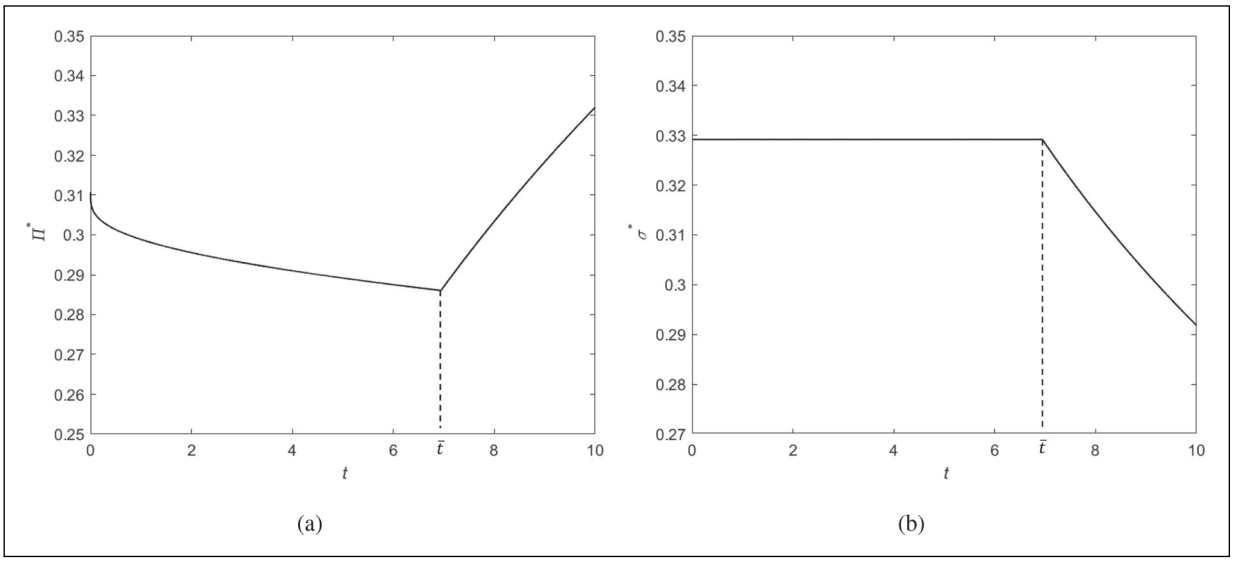


Figure 3. Authentic company's equilibrium profit and counterfeit probability; model parameters: $q = 10$, $p = 8$, $k = 2$, $L = 1$, $\alpha_1 = 0.2$, $\alpha_2 = 0.2$,$\gamma = 2$, and $n = 5$.

prohibitively high penalty $\left(L \geq \frac{p}{\alpha^\dagger} \min\left\{1, \frac{q}{p^\gamma}\right\}\right)$, which leads to the counterfeit-free case. The second one happens when the counterfeiter enters the market with certainty ($s^* = 1$). In this scenario, the penalty imposed on the counterfeiter is relatively low $\left(L < \frac{p}{\alpha^\dagger} \min\left\{1, \frac{q}{p^\gamma}\right\}\right)$ and the company's capability on anti-counterfeiting is high ($t > \bar{t}$). Accordingly, a slightly harsher penalty cannot disincentivize the counterfeiter from undertaking the opportunity of making and selling the counterfeits in such a lucrative market. Furthermore, note from equation (3) that an increase in penalty L does not have any direct impact on the authentic company's decision. As a result, the equilibrium counterfeit occurrence probability will not decrease with a marginal increase of the penalty. Note that from the definitions of equilibrium and σ^*, the customer demand $\int_0^1 \delta^*(\theta)d\theta$ can be rewritten as $\int_0^{\min\left\{q\left(\frac{1-\sigma^*}{p}\right)^\gamma, 1\right\}} d\theta$. Hence, in the above two scenarios, the increase of penalty level cannot attract more customers to buy a product on the market.

The comparative statics for the equilibrium anti-counterfeiting effort, on the contrary, is quite complicated. One may conjecture that an increase in the penalty may decrease the equilibrium anti-counterfeiting effort exerted by the authentic company, that is, e^* is decreasing in L, as there two measures serve the same purpose of preventing counterfeiting. However, Proposition 3(b) shows that such an argument may collapse in some circumstance. Specifically, when the penalty level L is low (i.e., $L < \left(\frac{\gamma}{(1+\gamma)p}\right)^\gamma \frac{qp}{\alpha^\dagger}$) and the authentic company's anti-counterfeiting capability is low (i.e., $t < \bar{t}$), the equilibrium anti-counterfeiting effort e^* is *increasing* in the penalty L (see Figure 4 for an illustration of regions with different penalty impacts).

To understand this result, we recall from Proposition 1 that in this case, we have $0 < s^* < 1$. In making the anti-counterfeiting effort decision, the authentic company needs to balance the marginal investment in its anti-counterfeiting technology and the marginal benefit. The impact of increasing penalty has two opposite effects on the company's profit. First, it can disincentivize the counterfeiter to enter the market, which can help save the company's anti-counterfeiting cost. On the other hand, increasing penalty can lower the counterfeit occurrence probability, which attracts more customers to purchase (see the expression of σ^* in Proposition 1). This effect, in turn, motivates the authentic company to exert more effort. Our result shows that when L is small and $0 < s^* < 1$, the latter effect is the dominant one.

The above result can yield an important and non-intuitive message to regulatory authorities and brand companies. Specifically, although governments of some emerging economies start to strengthen IPR enforcement (e.g., by increasing penalties imposed on counterfeiting activities), doing so probably cannot alleviate brand companies' cost burden brought by exerting anti-counterfeiting efforts, because those companies may not be able to enjoy a free ride by purely relying on the strengthened enforcement; instead, they need to increase their investments in anti-counterfeiting technologies as well. As a side proof, according to Nandi (2025), though law enforcements for intellectual property are continuously improved, the Asia-Pacific region is considered as the fastest growing market for anti-counterfeiting technologies.

We also conduct comparative statics analysis for model parameters α_1 and α_2, which are summarized as follows:

PROPOSITION 4. *Suppose* $L < \frac{p}{\alpha^\dagger} \min\left\{1, \frac{q}{p^\gamma}\right\}$. *The comparative statics results for* α_1 *and* α_2 *are:*

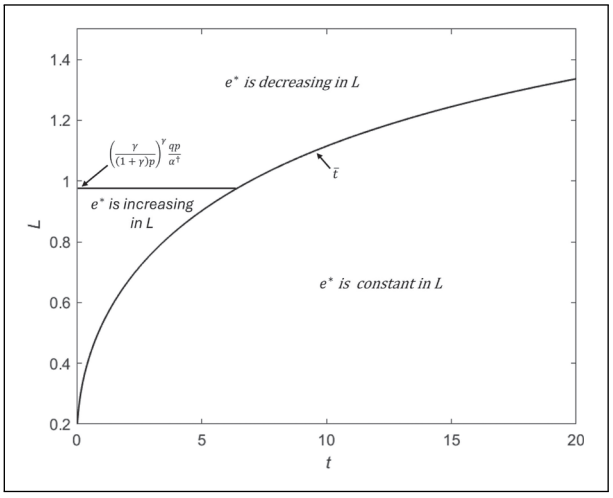


Figure 4. Impact of penalty on authentic company's anti-counterfeiting effort; model parameters: $q = 10$, $p = 8$, $k = 2$, $\alpha_1 = 0.2$, $\alpha_2 = 0.2$, $\gamma = 2$, and $n = 5$.

(a) *σ^* is decreasing in α_1 and α_2 for $t < \bar{t}$; σ^* is increasing in α_2 for $t > \bar{t}$.*

(b) *e^* is increasing in α_1 when $L < \left(\frac{\gamma}{(1+\gamma)p}\right)^{\gamma} \frac{qp}{\alpha^{\dagger}}$ and $t < \bar{t}$ and decreasing in α_1 when $L > \left(\frac{\gamma}{(1+\gamma)p}\right)^{\gamma} \frac{qp}{\alpha^{\dagger}}$ and $t < \bar{t}$; e^* is increasing in α_2 when $L < \left(\frac{(1-\alpha_1)(1-\alpha_2)\gamma}{[1+(1-\alpha_1)(1-\alpha_2)\gamma]p}\right)^{\gamma} \frac{qp}{\alpha^{\dagger}}$ and $t < \bar{t}$ and decreasing in α_2 when $t > \bar{t}$ or $t < \bar{t}$ and $L > \left(\frac{(1-\alpha_1)(1-\alpha_2)\gamma}{[1+(1-\alpha_1)(1-\alpha_2)\gamma]p}\right)^{\gamma} \frac{qp}{\alpha^{\dagger}}$.*

Proposition 4 first shows that the main comparative statics results for the penalty L are robust and can carry over to those for the enforcement intensities α_1 and α_2 when $t < \bar{t}$. For example, the equilibrium anti-counterfeiting effort e^* is quasi-concave in α_1 and α_2 and the equilibrium counterfeit occurrence probability is decreasing in α_1 and in α_2. The subtle difference is that the threshold on L for α_2, $\left(\frac{(1-\alpha_1)(1-\alpha_2)\gamma}{[1+(1-\alpha_1)(1-\alpha_2)\gamma]p}\right)^{\gamma} \frac{qp}{\alpha^{\dagger}}$, is lower than the threshold for α_1, $\left(\frac{\gamma}{(1+\gamma)p}\right)^{\gamma} \frac{qp}{\alpha^{\dagger}}$, indicating that the corrective legal action could be a more effective way to reduce the company's cost burden on anti-counterfeiting.

Proposition 4 further finds that when the anti-counterfeiting capability is relatively high, that is, $t > \bar{t}$, the equilibrium counterfeit occurrence probability is increasing in α_2. In other words, with a more stringent corrective law enforcement, it is more likely that counterfeit products appear on the market. In fact, due to compensation from corrective legal action and the fact that the counterfeiter is certain to enter the market, increasing the enforcement intensity makes the authentic company exert less anti-counterfeiting effort to protect its products from being imitated, resulting in a higher counterfeit occurrence probability.

5.2 Comparison Between Two Law Enforcements

We have incorporated two types of law enforcement into our model: preventive and corrective measures. As discussed, preventive law enforcement outlaws counterfeit goods before the counterfeiter can sell them to the market, while corrective law enforcement takes effect only after counterfeit goods are sold. We now examine how these two enforcement types influence the authentic company's equilibrium strategy and its equilibrium profit. To isolate their respective impacts, we consider two special cases. In the first case, only the preventive law enforcement is available (i.e., $\alpha_2 = 0$), while in the second case, only the corrective law enforcement is available (i.e., $\alpha_1 = 0$). To make a fair comparison, we assume that the probability of the law enforcement taking effect is identical across the two cases, which means $\alpha_1 = \alpha_2 = \alpha$. We use subscript 1 and 2 to present the equilibrium results for the case with only preventive law enforcement and that with only corrective enforcement, respectively. From the result of Proposition 1, it is clear to see that when $L \geq \frac{(1-\alpha)p}{\alpha} \min\left\{1, \frac{q}{p^{\gamma}}\right\}$, both the equilibrium e^* and the resulting counterfeit occurrence probability σ^* are equal to zero. Therefore, below we focus on the case when $L < \frac{(1-\alpha)p}{\alpha} \min\left\{1, \frac{q}{p^{\gamma}}\right\}$. We define

$$\bar{t}_1 = \frac{\psi'\left(\left[\frac{p}{1-\alpha}\left(\frac{1}{q}\frac{\alpha L}{(1-\alpha)p}\right)^{\frac{1}{\gamma}} - \frac{\alpha}{1-\alpha}\right]^{+}\right)}{(p-k)\frac{\alpha L}{p}}$$

and

$$\bar{t}_2 = \frac{\psi'\left(p\left(\frac{1}{q}\frac{\alpha L}{(1-\alpha)p}\right)^{\frac{1}{\gamma}}\right)}{(p-k)\frac{\alpha L}{p}},$$

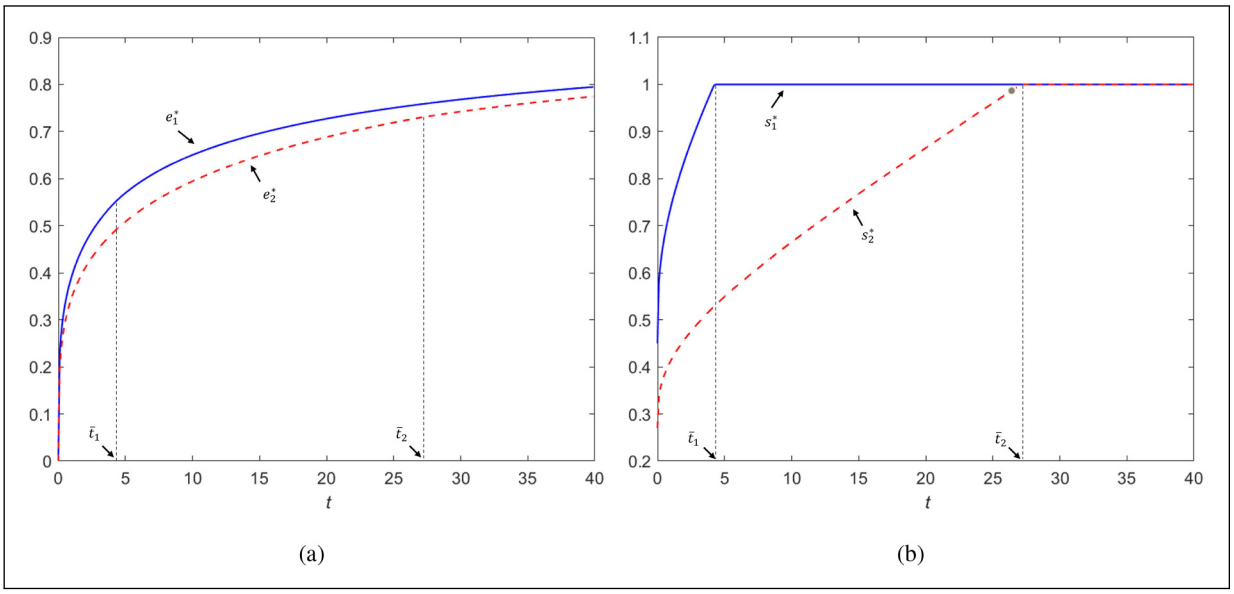


Figure 5. Equilibrium decisions with different law enforcements; model parameters: $q = 10$, $p = 8$, $k = 2$, $L = 1$, $\alpha = 0.4$, $\gamma = 2$, and $n = 5$.

where $\bar{t}_1 < \bar{t}_2$. The following proposition presents the comparison between preventive and corrective law enforcements.

PROPOSITION 5. *Suppose* $L < \frac{(1-\alpha)p}{\alpha} \min\left\{1, \frac{q}{p^\gamma}\right\}$. *The comparison of equilibrium solutions for the case with only preventive law enforcement and that with only corrective enforcement are summarized as follows:*

(a) $e_1^* \geq e_2^*$; $s_1^* > s_2^*$ *for* $t < \bar{t}_2$ *and* $s_1^* = s_2^*$ *otherwise.*
(b) $\sigma_1^* = \sigma_2^*$ *for* $t \leq \bar{t}_1$ *and* $\sigma_1^* < \sigma_2^*$ *for* $t > \bar{t}_1$.
(c) *There exists a* $t' \in [\bar{t}_1, \bar{t}_2]$ *such that* $\Pi_1^* < \Pi_2^*$ *for* $t < t'$ *and* $\Pi_1^* > \Pi_2^*$ *for* $t > t'$.

Proposition 5(a) shows that compared to corrective law enforcement, preventive law enforcement induces higher effort from the authentic company and makes the counterfeiter entry more likely (see Figure 5(a) and (b) for illustration). This result is mainly due to the increased customers' expectation on the genuineness of the product to be purchased. Indeed, when the preventive law enforcement is in place, the products sold on the market are more likely to be the genuine ones since the counterfeit products are likely to be outlawed before the purchase.

Proposition 5(b) and (c) shows that the two types of law enforcement pose different impacts on the authentic company's profit and the probability of counterfeit occurrence. Specifically, compared to the corrective law enforcement, when the authentic company's capability of anti-counterfeiting is high, the preventive law enforcement not only lowers the equilibrium counterfeit occurrence probability (thus, attracting more customers to purchase) but also leads to a higher equilibrium profit for the authentic company. However, when the capability of anti-counterfeiting is low, the preventive law enforcement would make the authentic company *worse off* (see Figure 6). The reason behind this counter-intuitive result is as follows. Note that when the preventive law enforcement is in place, from Proposition 5(a), the counterfeiter becomes more aggressive (via setting a higher s_1^*) when it can imitate the genuine products. Because the counterfeiter is more likely to enter the market, the authentic company is forced to exert more anti-counterfeiting effort as well, thus increasing the company's cost burden. Unfortunately, the customers cannot observe this increased effort due to the lack of domain knowledge on anti-counterfeiting. Thus, to the authentic company, the benefit brought by this increased effort can hardly compensate for its cost spent in equilibrium, leading to an over-investment problem.

In short, our analysis highlights the value of brand companies' collaboration with regulatory authorities and more importantly, implies that the value depends heavily on the coupling between the anti-counterfeiting capability and the law enforcement type. Particularly, when a brand company has a good anti-counterfeiting capability, the company needs to promote a preventative collaboration with the regulatory authority so that the regulatory authority puts more resources into preventive law enforcement, such as inspecting factory floors and warehouses of local business owners that make similar products. That way, the counterfeits of the company's products could hardly be produced and delivered to the market. On the other hand, if the company has a weak anti-counterfeiting capability, it is desirable for the company to establish a reactive collaboration with the authority by which the authority

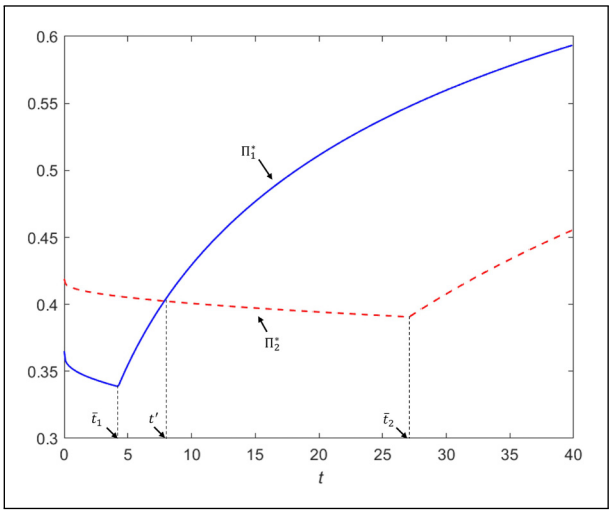


Figure 6. Equilibrium profits with different law enforcements; model parameters: $q = 10$, $p = 8$, $k = 2$, $L = 1$, $\alpha = 0.4$, $\gamma = 2$, and $n = 5$.

will place more focus on handling infringement reports during its selling process, such as collecting evidence from the market, conducting legal prosecutions against counterfeiters, and compensating the brand company with levied fines. A blinded cooperation without considering its own capability of anti-counterfeiting may not serve the company's best interest, yielding an over-protection of the company's brand and a decrease in its profit.

Finally, we analyze how these two types of law enforcement affects the consumer welfare. To that end, we follow the literature and define the consumer surplus CS as the aggregate consumer utilities. Specifically, from the customer's utility defined in Section 3, we can write the equilibrium consumer surplus as follows:

$$CS^* = \int_0^1 \left(b^* \left(\frac{q}{\theta} \right)^{\frac{1}{\gamma}} - p \right) \delta^*(\theta)\, d\theta. \tag{6}$$

With the result of Proposition 1, CS^* can be expressed as follows:

$$CS^* = \begin{cases} \dfrac{\alpha^\dagger L}{\gamma - 1} & \text{if } t < \bar{t}, \\ \min\left\{ q \left(\dfrac{e^* + (1 - e^*)\alpha_1}{p} \right)^\gamma, 1 \right\} \left(\dfrac{\gamma}{\gamma - 1} \dfrac{q^{\frac{1}{\gamma}} (e^* + (1 - e^*)\alpha_1)}{\min\left\{ \frac{q^{\frac{1}{\gamma}} (e^* + (1 - e^*)\alpha_1)}{p}, 1 \right\}} - p \right) & \text{if } t \geq \bar{t}, \end{cases} \tag{7}$$

We also define CS_1^* (and CS_2^*, respectively) as the equilibrium consumer surplus when only the preventive (and corrective, respectively) law enforcement is available. The impacts of the law enforcements on consumer surplus can be summarized in the following proposition:

PROPOSITION 6. *Suppose* $L < \frac{(1-\alpha)p}{\alpha} \min\left\{1, \frac{q}{p^\gamma}\right\}$. *The following holds:*

(a) CS^* *is increasing in* L *and constant in* t *when* $t < \bar{t}$ *but constant in* L *and increasing in* t *when* $t > \bar{t}$.
(b) $CS_1^* = CS_2^*$ *for* $t \leq \bar{t}_1$ *and* $CS_1^* > CS_2^*$ *for* $t > \bar{t}_1$.

Proposition 6(a) indicates that increasing the penalty will not improve the consumer welfare when $t > \bar{t}$. This is because the counterfeiter enters the market with certainty, which is consistent with the discussion that follows Proposition 2. In addition, improving the anti-counterfeiting capability cannot improve the consumer welfare when $t < \bar{t}$. In this case, improving the anti-counterfeiting capability makes the counterfeiter more likely to produce and sell a counterfeit (if its realized product cost is low), thereby completely negating its potential benefit to customers.

Proposition 6(b) shows when $t > \bar{t}_1$, the preventative law enforcement yields a higher consumer surplus than the

corrective law enforcement. In that case, as shown in Proposition 5(b), the preventative law enforcement is more effective in reducing the counterfeit occurrence probability. Also note from Proposition 5(c) that $\Pi_1^* < \Pi_2^*$ when $t < t'$ and $\Pi_1^* > \Pi_2^*$ when $t > t'$. Therefore, our analytical results indicate that with a strong anti-counterfeiting capability, allocating more resources to the preventative law enforcement benefits both brand companies and customers, which underscores the importance of promoting an effective cooperation with the regulatory authority.

6 Model Extensions and Discussions

In this section, we present several extensions of the base model. In Section 6.1, we take into account a positive product cost for the counterfeiter and discuss its impact on the equilibrium solution. Next, we explore the adverse impact of selling counterfeit products on the authentic company in Section 6.2 and study the case that the company has the commitment ability to its anti-counterfeiting effort in Section 6.3. Finally, we model the authentic company's pricing decision endogenously in Section 6.4. All figures in this section can be found in EC.4.

6.1 Product Cost of the Counterfeiter

We extend our analysis to the case of a positive product cost, $c_l > 0$ for the counterfeiter when the anti-counterfeiting technology turns out to be unreliable. To differentiate the notation used in this case from that in the base model, we denote the decision variables with a "hat." Proposition 7 demonstrates how the positive product cost for the counterfeiter impacts the equilibrium decisions and the counterfeit occurrence probability.

PROPOSITION 7. *When* $L \geq \frac{(1-\alpha_1)(1-\alpha_2)p-(1-\alpha_1)c_l}{\alpha_1+(1-\alpha_1)\alpha_2} \min\left\{1, \frac{q}{p^\gamma}\right\}$, *the equilibrium anti-counterfeiting effort is* $\hat{e}^* = 0$, *and the counterfeiter's equilibrium effort is* $\hat{s}^* = 0$. *When* $L < \frac{(1-\alpha_1)(1-\alpha_2)p-(1-\alpha_1)c_l}{\alpha_1+(1-\alpha_1)\alpha_2} \min\{1, \frac{q}{p^\gamma}\}$, *there exists a threshold* $\hat{t}$ *such that* $0 < \hat{s}^* < 1$ *if* $t < \hat{t}$ *and* $\hat{s}^* = 1$ *otherwise. Moreover, when* $t < \hat{t}$, $\hat{e}^*$ *is quasi-concave in* c_l, *and* $\hat{\sigma}^*$ *are decreasing in* c_l; *when* $t \geq \hat{t}$, *both* $\hat{e}^*$ *and* $\hat{\sigma}^*$ *are constant in* c_l.

Proposition 7 shows that our main results in the base model are robust. Specifically, the counterfeiter enters the market only when the penalty L is low; the counterfeiting effort $\hat{s}^* = 1$ if and only if the anti-counterfeiting capability is greater than a threshold, $\hat{t}$.

Proposition 7 further reveal some interesting insights. In particular, a higher c_l makes it more costly for the counterfeiter to produce a fake and thus leads to a lower counterfeit occurrence probability, presumably reducing the authentic company's anti-counterfeiting burden. However, as shown in Proposition 7 and Figure EC.1, the anti-counterfeiting effort $\hat{e}^*$ could be non-monotone in c_l for $t < \hat{t}$. In that case, there are two forces working against one another. On the one hand, as a higher c_l makes it more costly for the counterfeiter to engage in the counterfeiting business, it reduces the authentic company's incentive to exert its anti-counterfeiting effort. On the other hand, a higher c_l attracts more customers to make a purchase, which increases the company's incentive to exert the anti-counterfeiting effort. When the product cost c_l is low (high, respectively), the second (first, respectively) force dominates, and the equilibrium anti-counterfeiting effort $\hat{e}^*$ is increasing (decreasing, respectively) in c_l.

6.2 Impact of Selling Counterfeits

In the base model, we assumed that the sales of the genuine products drop to zero if the counterfeiter is able to sell the counterfeits on the market. In this section, we extend the model by considering the case in which the availability of counterfeits may have a limited impact on the sales of the genuine products and use a parameter β to capture the impact. Specifically, with probability β, a customer will encounter a counterfeit product; with probability $1-\beta$, a customer are still able to buy the genuine one (although they are unaware of the genuineness of the product at the time of purchase). Thus, $\beta = 0$ is the case in which all the customers can purchase a genuine one even in the presence of the counterfeits while $\beta = 1$ represents the base model studied previously. We add a "tilde" to the notation used this extension. The following proposition presents how the model parameter β affects the equilibrium decisions.

PROPOSITION 8. *The counterfeiter's effort* $\tilde{s}^* > 0$ *if and only if* $L < \frac{\beta p}{\alpha^\dagger} \min\left\{1, \frac{q}{p^\gamma}\right\}$. *Under* $L < \frac{\beta p}{\alpha^\dagger} \min\left\{1, \frac{q}{p^\gamma}\right\}$, *there exists a threshold* $\tilde{t}$ *such that* $\tilde{s}^* = 1$ *if* $t > \tilde{t}$ *and* $0 < \tilde{s}^* < 1$ *otherwise. Moreover,* $\tilde{e}^*$ *is decreasing in* β *when* $L < \frac{\beta p}{\alpha^\dagger} \min\left\{1, q\left(\frac{\gamma}{\gamma+1}\frac{1}{p}\right)^\gamma\right\}$ *and* $t < \tilde{t}$.

As shown in Proposition 8, our analytical results in the base model continue to hold for this extension. Proposition 8 further shows that the equilibrium anti-counterfeiting effort $\tilde{e}^*$ could be decreasing in β. Note that β is the probability that a customer encounters a counterfeit product and thus represents the impact of the counterfeits on the sales of the genuine products. Thus, as β increases, the impact of counterfeiting becomes more severe, but it discourages the authentic company from exerting its anti-counterfeiting effort. This non-intuitive result comes again from the customers' reaction. In this case, the effect of low customer demand caused by the product's reduced market attractiveness outweighs the market share taken by the counterfeiter. As a result, the company exerts a lower anti-counterfeiting effort in equilibrium.

6.3 Commitment to Anti-counterfeiting Effort

In the previous analysis, we have assumed that the customers have no information regarding the authentic company's anti-counterfeiting effort. In reality, some brand companies can

commit to their anti-counterfeiting efforts by publicizing or signaling their efforts credibly.[12] In this extension, we study a setting in which the authentic company can credibly commit to its anti-counterfeiting effort. Our aim is to evaluate how this commitment ability influences the authentic company's decision and the resulting counterfeit occurrence probability. For the purpose of differentiation, we add "check" to the equilibrium results in this extension.

Unlike in the base model, a customer's belief now depends on the anti-counterfeiting effort $\check{e}$ exerted by the authentic company. Consequently, given $\check{e}$, the counterfeiter's effort and the customer's belief are functions of $\check{e}$, which we denote by $\check{s}^*(\check{e})$ and $\check{b}^*(\check{e})$, respectively. Then, when the authentic company can commit to its anti-counterfeiting effort, its decision-making problem is

$$\max_{0\le\check{e}\le 1}\left\{\begin{array}{l}\left[\check{e}+(1-\check{e})\left(1-\check{s}^*(\check{e})+\check{s}^*(\check{e})\alpha_1\right)\right](p-k)\displaystyle\int_0^{\min\left\{q\left(\frac{\check{b}^*(\check{e})}{p}\right)^{\gamma},1\right\}}d\theta\\ +(1-\check{e})\,\check{s}^*(\check{e})\left(1-\alpha_1\right)\alpha_2(p-k)\displaystyle\int_0^{\min\left\{q\left(\frac{\check{b}^*(\check{e})}{p}\right)^{\gamma},1\right\}}d\theta-\frac{1}{t}\psi\left(\check{e}\right)\end{array}\right\},$$

where the formulas for $\check{s}^*(\check{e})$ and $\check{b}^*(\check{e})$ are presented in detail in EC.3. Solving the above maximization problem yields the authentic company's optimal strategy. We present the main findings in the following proposition:

PROPOSITION 9. *When* $L \ge \frac{p}{\alpha^\dagger}\min\left\{\frac{q}{p^\gamma},1\right\}$, $\check{e}^*=0$. *When* $L<\frac{p}{\alpha^\dagger}\min\left\{\frac{q}{p^\gamma},1\right\}$, *there exists* $\check{t}\ge 0$ *such that* $\check{e}^*=0<e^*$ *if* $t<\check{t}$ *and* $\check{e}^*\ge e^*$ *otherwise. Moreover,* $\check{\Pi}^*\ge\Pi^*$; $\check{\sigma}^*>\sigma^*$ *if* $\bar{t}<t<\check{t}$ *and* $\check{\sigma}^*\le\sigma^*$ *otherwise.*

Proposition 9 demonstrates how the authentic company's ability to commit to its anti-counterfeiting effort may influence the level of effort exerted (see Figure EC.2(a) for an illustration). When the anti-counterfeiting capability is low, that is, $t<\check{t}$, the authentic company may choose not to exert any effort, resulting in $\check{e}^*=0<e^*$. In this case, exerting anti-counterfeiting effort is costly for the authentic company and will motivate the counterfeiter to respond by setting a higher s, thus lowering the authentic company's profit. Therefore, the company would be better off with $\check{e}=0$. When the anti-counterfeiting capability is high, that is, $t>\check{t}$, the ability to commit the anti-counterfeiting effort yields a higher effort level compared to that in the base model without such commitment ability. The reason comes from the fact that the commitment allows the authentic company to internalize the customers' positive response to the value brought by the anti-counterfeiting effort. In addition, as illustrated in Figure EC.2(b), committing to the effort can always benefit the company, since its decision, once committed, can effectively influence the counterfeiter's and customers' decisions.

Proposition 9 further shows that possessing the commitment ability does not necessarily help the authentic company reduce the counterfeit occurrence probability. When the anti-counterfeiting capability t is moderate, this probability can actually *increase* (see Figure EC.3 for an illustration). The reason for this interesting result is as follows: to maximize its profit, the authentic company needs to strike a balance between the revenue gained from the market and the cost of adopting any anti-counterfeiting technology. When $t<\check{t}$, the anti-counterfeiting capability is insufficient. Consequently, the authentic company would save its expenditure on anti-counterfeiting by reducing its anti-counterfeiting effort to zero when it can commit to this decision. This commitment ability, while saving the cost of anti-counterfeiting effort, may provide the counterfeiter with more opportunities on the market, leading to a higher counterfeit occurrence probability. In short, although the commitment may improve the authentic company's profitability, it could adversely increase the counterfeit occurrence probability, making its products less attractive to customers.

6.4 Pricing Decision

In the base model, we have assumed that the market price p is exogenous. In this subsection, we investigate how our results are impacted when the authentic company can set the price. With p as a decision variable for the authentic company, we note that the new sequence of events is the same as that presented in the base model except that the company needs to decide on the price at the beginning of the game.[13] Given the complexity of the problem, we first present some theoretical observations regarding the price, followed by a numerical study to explore its properties.

We begin the discussion with a counterfeit-free case in which there is no counterfeiter on the market and denote the authentic company's optimal price in that case by p^0. We can solve for the optimal price analytically and obtain

$$p^0=\max\left\{\frac{\gamma k}{\gamma-1},q^{\frac{1}{\gamma}}\right\}.$$

As is clear, due to the absence of the counterfeiter, the authentic company does not need to exert any anti-counterfeiting effort, while the law enforcement plays no role. Thus, the optimal price in the counterfeit-free case depends only on the product's characteristics, including the quality q, the unit cost k, and the constant elasticity γ.

When the counterfeiter is present, for the exogenous price considered in the base model, Proposition 1 reveals an important relationship between the counterfeiter's equilibrium decision s^* and the given price p, which we highlight in the following corollary:

COROLLARY 1. *$s^* > 0$ if and only if $\alpha^\dagger L \le p \le (\frac{q}{\alpha^\dagger L})^{\frac{1}{\gamma-1}}$.*

Note that the counterfeiter's decision s is an indicator of its market entry, with $s > 0$ ($s = 0$, respectively) representing an entry (no entry, respectively). Thus, the above corollary suggests that the counterfeiter enters the market only when the given price on the market is moderate. This result comes from the fact that with a very low (high, respectively) market price, the counterfeiter cannot gain an adequate profit margin (demand, respectively) to offset the probable penalty levied by the regulatory authority. As will be presented shortly, such observation will help us understand the authentic company's pricing strategy when it can set its price freely.

The above corollary can serve as a bridge to analyzing the case in which p is endogenized as a decision variable. In that case, if the counterfeiter's decision is $s^* = 0$ (i.e., no entry) in equilibrium, then the authentic company's equilibrium decision must be the same as that in the counterfeit-free case. Thus, we can derive from the corollary a sufficient condition for the authentic company to set its product price the same as p^0 as follows (note that $\alpha^\dagger L \le p \le \left(\frac{q}{\alpha^\dagger L}\right)^{\frac{1}{\gamma-1}}$, as stated in the corollary, becomes vacuous under the sufficient condition):

$$q < \left(\alpha^\dagger L\right)^\gamma .$$

The sufficient condition indicates when the product quality is relatively low and/or the penalty from the regulatory authority is relatively stringent, the counterfeiter is better off with not entering the market due to the low profit margin and/or small demand. Consequently, the authentic company would set its product price at p^0, the optimal one for the counterfeit-free case.

We now turn to general cases (especially the case in which the sufficient condition collapses). Given that our research focuses lies in the authentic company's decision on exerting the anti-counterfeiting effort, we are particularly interested in how the company's decision interacts with its anti-counterfeiting capability. However, we find that it is no longer tractable to derive the equilibrium solution in closed form (the technical challenge stems from the fact that, the price p appears not only in the expression of the authentic company's profit function Π^* but also in the expression of threshold $\bar{t}$, as illustrated in Proposition 1). Thus, in what follows, we will present an asymptotic result for the authentic company's optimal pricing relative to its anti-counterfeiting capability, followed by a numerical study to explore extra insights.

PROPOSITION 10. *$p^* \to p^0$ as $t \to +\infty$.*

The asymptotic behavior of the equilibrium price p^* is intuitive. Specifically, as the authentic company's anti-counterfeiting capability approaches perfection, the counterfeits becomes very unlikely to appear on the market. As such, the market resembles one in the counterfeit-free case, leading to a convergence of the equilibrium price p^* to p^0, the optimal price for the counterfeit-free case.

We then conduct a numerical experiment to study how the authentic company's equilibrium decision on the price varies with its anti-counterfeiting capability. The result is presented in Figure EC.4. A straightforward observation from the figure is that the equilibrium price p^* has a tendency to converge to the optimal price p^0 in the counterfeit-free case, which is consistent with what is stated in Proposition 10. More importantly, Figure EC.4 demonstrates how the existence of counterfeits may cause a departure of the optimal price from p^0, the one in the counterfeit-free case. Specifically, we can see that the price can be either higher or lower than p^0, depending on the company's anti-counterfeiting capability. When the authentic company's anti-counterfeiting capability is relatively low, the company is more inclined to adopt a high-price strategy. Given the fact that the counterfeiter has to set the same price for its deceptive counterfeits, such strategy essentially shrinks the demand available to the counterfeiter. Thus, in the presence of the probable penalty to be levied by the regulatory authority, the total profit from engaging in the counterfeiting business is squeezed, making it more reluctant to enter into the market. On the contrary, when the company's anti-counterfeiting capability is relatively high, it becomes preferable for the company to employ a low-price strategy. As noted in Proposition 1, in this case, the counterfeiter is more likely to enter the market when the anti-counterfeiting technology proves unreliable. Consequently, setting a higher price is no longer effective in shrinking the demand available to the counterfeiter. Thus, the company should adopt a low-price strategy instead. That way, the company can reduce the unit profit margin for the counterfeiter, which can discourage the latter from making and selling counterfeits.

7 Concluding Remarks

In this article, we consider a market in which an authentic company aims to sell genuine products while facing the potential imitation by a counterfeiter. To counter the probable counterfeiting activity, the authentic company can develop an anti-counterfeiting technology with uncertain outcomes. At the same time, the market is administered by a regulatory authority, who may deploy an IPR team to outlaw the fake

through preventive and/or corrective measures. We employ a game-theoretic model to analyze the problem and derive the equilibrium solution, from which we obtain several important insights.

We examine how the authentic company's equilibrium profit may vary with its anti-counterfeiting capability. Interestingly, when the capability is relatively low, an increase in its capability would lead to a reduction in the equilibrium profit. This result implies that if the authentic company wants to target high profitability, it may have to forgo an improvement on this capability unless there exists a substantial opportunity to enhance the capability significantly.

Second, we explore the interaction between the two widely used approaches to combating counterfeits, the anti-counterfeiting technology adopted by the authentic company and the penalty levied by the regulatory authority. While those two approaches serve the same purpose of preventing and combating counterfeits, our analysis indicates that the anti-counterfeiting effort exerted by the authentic company does not necessarily decrease in the penalty levied by the authority. In particular, we show that when the penalty is low, an increase in the penalty (which is likely to be the case in the market of emerging economies) may be accompanied by an increased investment in anti-counterfeiting technology. We also find that a stringent law enforcement may lead to a higher counterfeit occurrence probability.

Third, we examine how preventive and corrective law enforcement may influence the authentic company differently. Interestingly, we find that preventive law enforcement can more effectively reduce the counterfeit occurrence probability but lead to a lower profit of the authentic company. This happens when the anti-counterfeiting capability is relatively low. Thus, our analysis implies the importance of the coupling between anti-counterfeiting investment and law enforcement type when the authentic company seeks a cooperation with the regulatory authority. Moreover, we examine the consumer welfare implications of these two law enforcement types. Our result shows that when a strong anti-counterfeiting capability is in place, preventative law enforcement benefits not only brands but also customers.

Finally, we extend our base model by incorporating several additional features. Results show that our insights in the base model are robust. These extensions also have yielded some non-intuitive insights. For example, the authentic company's equilibrium anti-counterfeiting effort could be non-monotone in the counterfeit product cost, and a greater adverse impact from selling counterfeits might not result in a higher anti-counterfeiting effort in equilibrium. Furthermore, committing to its anti-counterfeiting effort could adversely increase the counterfeit occurrence probability. These results collectively suggest that combating counterfeit products is inherently complicated, involving careful examination and planning.

We suggest some avenues for future research. In our model, we have assumed that customers cannot tell the genuineness of a product and has to form a belief when making the purchase decision (unless the authentic company can credibly commit to its effort). We can extend the analysis by considering another setting in which customers are partially informed. For example, customers with some research may be able to observe the coexistence of genuine products and fake ones in the market. It would be interesting to study how the partial information may influence the customer's expectation and accordingly, the equilibrium decisions made by the authentic company and the counterfeiter. The second avenue for extending the analysis is to incorporate the authentic company's endogenous effort to detect and report counterfeiting activities, capitalizing on their expertise in identifying misconduct (Babich and Tang, 2012). It would be valuable to explore how the authentic company's such efforts impact the development of their anti-counterfeiting technology. The third direction to extend the analysis is to consider a case where the regulatory authority is endogenized as a game player (e.g., Pun et al., 2021). In that case, the authority needs to consider the trade-off between its efforts put into regulating the market and the social benefit gained from outlawing counterfeits. It is desirable to study which type of law enforcement, as well as what level of enforcement effort, is the optimal choice for the regulatory authority. Finally, in our model, we have assumed that the counterfeiter sources imitation labels from external suppliers via illicit channels. We can extend the analysis to the case where the counterfeiter can actively exert its own effort to imitate the anti-counterfeiting features developed by the authentic company, rather than relying on external sources. To model this extension, we think that a sequential contest framework from the economics literature could provide a useful approach, as it allows for an analysis of the strategic interaction between the authentic company and the counterfeiter, taking into account the technological competition and the dynamics of imitation.

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Notes

1. See https://www.cbp.gov/trade/priority-issues/ipr. (Retrieved Dec 10, 2023)
2. See https://www.marketsandmarkets.com/Market-Reports/anti-counterfeit-packaging-advanced-technologies-and-global-market-129.html. (Retrieved Dec 10, 2023)
3. To focus on company's non-pricing anti-counterfeiting measures, we assume that the price is predetermined in the base model, while extending the analysis by considering the endogenous pricing decision in Section 6.4; see similar assumptions in the recent literature, for example, Guo et al. (2022), Lu et al. (2023), Plambeck and Taylor (2016), and Zhou et al. (2024). This assumption is also consistent with the practice by companies in certain industries that enter developing economies (Danzon et al., 2000; Irwin et al., 2004). The most prominent example is pharmaceutical companies. Developing countries such as India and Brazil have very stringent price regulation policies that prevent foreign pharmaceutical companies from freely setting prices of essential medicines. India's Drug Price Control Order (DPCO) is an exemplary regulation that limits prices of over 800 formulations, ensuring affordability of the medicine to those in need (Kolte et al., 2021).
4. Note that the customer utility function can be equivalently written as $\left(\frac{1}{\theta}\right)^{\frac{1}{\gamma}} q^{\frac{1}{\gamma}} - p$, where the coefficient $\left(\frac{1}{\theta}\right)^{\frac{1}{\gamma}}$ can be interpreted as consumer wealth level (Gao et al., 2017; Rao and Schaefer, 2013; Yayla-Küllü et al., 2021). Given that θ follows a uniform distribution, the coefficient $(\frac{1}{\theta})^{\frac{1}{\gamma}}$ follows the Pareto distribution (a power-law probability distribution) with shape parameter γ and scale parameter 1, which has a probability density function of $\gamma x^{-(1+\gamma)}$ for $x > 1$ and 0 for $x < 1$. This distribution is positively skewed, with a long right tail that implies only a small portion of consumers have disproportionately high valuations (or willingness to pay) for product quality. In the literature, probability distributions with long right tails can capture empirical patterns observed in wealth and income distributions (Angrist and Pischke, 2009; Bureau, 2015). Particularly, Gabaix (2009) summarizes several studies that report wealth and income distributions exhibiting power-law (Pareto) behavior.
5. Most packaging-related anti-counterfeiting technologies operate as stand-alone systems (Soon and Manning, 2019). Customers often have a difficulty in differentiating a genuine product from a fake one that has exactly the same set of anti-counterfeiting features. Our model can be extended to the case where there are two types of customers, expert customers and non-expert customers. Expert customers can always identify the counterfeits while the non-expert customers cannot distinguish the counterfeits from the genuine ones. Since the authentic company's profit from expert customers is constant, our main results do not change except that the customer size available to both the authentic company and the counterfeiter is now the size of non-expert customers.
6. Counterfeiters may opt to procure imitated anti-counterfeiting features from the black markets rather than develop them independently. This is because their core competencies often lie in producing counterfeit goods and establishing distribution networks. They may lack the technological capabilities to imitate anti-counterfeiting features developed by authentic companies. Thus, relying on the black market becomes a viable option for counterfeiters lacking such capabilities.
7. To isolate the impact of anti-counterfeiting technology on preventing label imitation, we assume that the manufacturing cost of the counterfeit product is independent of the reliability outcome.
8. It suffices to assume that $c_h > p$, as p is the maximum profit the counterfeiter can extract from the market for each unit.
9. Note that the fixed expenses associated with anti-counterfeiting are generally significant relative to the costs per unit. In this article, we simplify our model by assuming the per-unit cost is zero (see, for instance, Gao, 2018 for the similar assumption).
10. We assume that n is larger than $2 + \gamma$ to ensure sufficient convexity of the cost function $\psi(e)$ for equilibrium uniqueness.
11. In Section 6.3, we consider the case in which the authentic company's effort can be observed by the customers.
12. For example, Brother Industries, Ltd., a Japanese multinational electronics and electrical equipment company, announces the introduction of a revolutionary holographic security label that provides visual authentication and online authentication, helping customers to identity the genuineness of a product (Brother, 2017).
13. It is worth mentioning that the order in which the price and the anti-counterfeiting effort are determined has little impact on the equilibrium outcome. This is because both decisions are made by the authentic company before any actions by the other players.

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